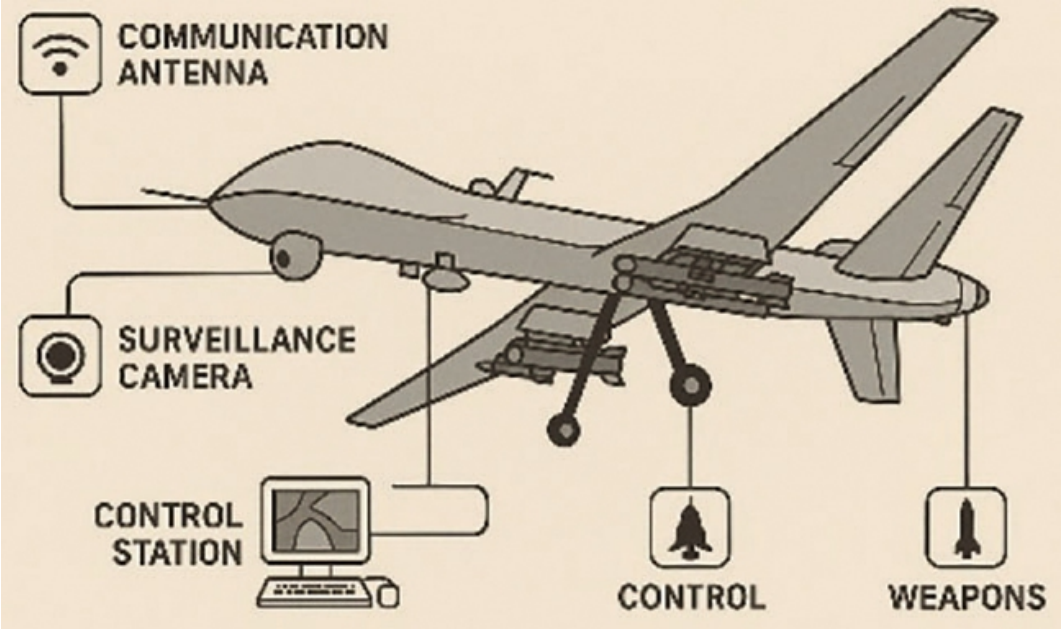


Figure 1.1 Strategic utilization of drones in modern warfare. ⏎

1.2.2 Commercial and civilian use

In the civilian and industrial sectors, drone technology has become synonymous with efficiency, safety, and innovation. Industries leverage UAVs for real-time data collection, precision mapping, and infrastructure monitoring. In agriculture, drones equipped with multispectral and hyperspectral sensors assist in yield estimation, irrigation planning, and pest management, leading to improved productivity and resource optimization. In construction and infrastructure, UAVs are revolutionizing surveying, inspection of bridges and pipelines, and project monitoring, reducing both cost and human risk.

Environmental and emergency management applications also benefit immensely from drones, assisting in wildfire detection, flood monitoring, and disaster relief by providing rapid situational data and enabling the delivery of essential supplies to inaccessible areas. In logistics, companies such as Amazon and Zipline are pioneering drone-based delivery systems that promise faster, contactless transportation of goods, medical supplies,